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CASE REPORT

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An experience-based model on how reflection may facilitate and foster professionals' learning in lifelong learning initiatives

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ABSTRACT

Today's industrial professionals encounter numerous emerging digital solutions, in the office, as well as, on the shop floor. While these solutions present significant opportunities, they also introduce challenges that must be addressed. A key part of these challenges is often a lack of up-to-date knowledge, where an educational degree, obtained years past, is no longer sufficient. Acquiring new knowledge is essential, yet, when combined with ongoing work obligations, competence development initiatives require a different approach and structure compared to traditional full-time, on-campus education. Consequently, alternative strategies are necessary, and the concept of lifelong learning is currently attracting considerable attention from both policymakers and academia. A central question in this context is: how can individuals acquire new knowledge, while at the same time continuing to contribute value in their current professional roles? Therefore, this paper addresses two related questions: How can online courses be designed and delivered to enable industry professionals to combine full-time work with lifelong learning initiatives? And how can such courses enhance professionals' ability to transform newly acquired theoretical knowledge into practical, internalised skills applicable in their daily work? We present an experience-based model for the design and execution of a general course using an asynchronous approach.

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Lifelong learning; reflective practitioner; online course design; course for professionals; internalising knowledge

Introduction

Today's rapid technological development permeates all sectors, influencing business models and products, as well as how day-to-day work routines are carried out. Everyone is experiencing rapid change, and professionals in technology-intensive fields are no exception; new knowledge must continuously be acquired. However, it is not realistic to expect these professionals, who are often in mid-life, balancing family responsibilities, housing mortgages, and active social lives, to leave their jobs for extended

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periods to pursue an additional university degree. Instead, continuous competence development needs to be integrated as a natural part of their professional lives.

In this paper, we address colleagues at universities who are involved in lifelong learning initiatives for full-time working professionals. Our work focuses on two related questions. First, we examine how to design and deliver courses that enable professionals to combine full-time work with lifelong learning. The ability to participate is fundamental to developing one's knowledge. Second, we explore how to enhance professionals' understanding of the connection between the theories and methodologies presented in the course and their current practice, with the aim of promoting active learning through reflection and assessment as learning activities.

We present an approach for designing and delivering online courses for lifelong learning and discuss experiences from cases involving mid-career professionals. The courses are situated in a Swedish context and, in accordance with national regulations, professionals are able to attend these university-level courses free of charge. To concretise the structure and implementation, data and examples from five online courses conducted between 2020 and 2025 are used as the primary empirical material in this study (Table 1). All five courses were conducted fully online, without onsite meetings, and relied on

Table 1. The online courses for professionals presented in this paper.

Column A – Number of registered professionals on the course
Column B – Number of active professionals after one month into the course (– no data available)
Column C – Number of professionals who passed all examinations

Year	A	B	C	*Name* (ECTS-credits; Grading) – Level – Course content and key assessment methods (*ECTS – European Credit and Accumulation System)*
2020	57	-25	24	*Technical Leadership for Industry in Change* (3 ECTS-credit; Pass/Fail) – Undergraduate level –
2021	39	23	14	Aimed at professionals in industry who have no previous experience of management tasks
2022	36	30	18	or who are new to their role as a manager and who wish to gain basic knowledge that will
2023	46		14	support them on their way into such an assignment. The course gives introduction to the field of technical leadership. Assessment methods used are written report and presentation/ reflection of leadership based in own experience.
2020	24	19	13	*Technical Leadership for the Industrial Processes of the Future* (3 ECTS-credit; Pass/Fail) – Master
2021	6	5	5	level – The course is designed for professionals in industry who have, or strive to have, a leading role in an industrial organisation. It provides further knowledge in the field of technical leadership. Assessment methods used are written report and presentation/ reflection leadership based in own experience.
2022	30	24	12	*Technologies in Industry 4.0* (3 ECTS-credit; Pass/Fail) – Master level – The course summarises
2023	23	13	8	Industry 4.0 technologies and focuses on how these, often disruptive technologies can affect companies, society and the environment from a wider perspective. Assessment methods used are written report and presentation/reflection of how emerging technology could be implemented at work.
2022	17	15	8	*Industrial Digital Transformation* (3 ECTS-credit; Pass/Fail) – Master level – The course
2023	26	21	9	complements a strong technological association with a number of other important business dimensions (e.g. strategy, culture, employees, customers). The importance and impact of the different dimensions on each other and the opportunities that new technical solutions and digitalisation may initiate is discussed. Assessment methods used are written report and presentation/reflection of how transformation can be applied at work.
2024	67	49	29	*Technical Leadership in a Digital Transformation* (7.5 ECTS-credit; Pass with distinction/Pass/
2025	104	74	53	Fail) – Undergraduate level – The course positions itself in the intersection between the subject area's technical leadership and digital transformation. It thereby focuses on the influence and importance of technical leadership in a digital transformation, as well as how a digital transformation alters the prerequisites of and the competences needed in a technical leader. Assessment methods used are quiz, written report and presentation/ reflection of how participants could manage transformation at work.

recorded lectures. While originally designed for professionals in the manufacturing industry, professionals from other sectors also attended.

The courses have been designed to foster active learning through reflection, enabling professionals to internalise new knowledge by integrating theories and methods introduced in the course with their prior experience (Jaiswal et al., 2021). The design and execution are grounded in Kolb's learning cycle (D. A. Kolb, 1984, 2007; A. Y. Kolb & Kolb, 2009), following its four stages.

Experiential learning, or 'learning by doing' is, according to Dewey (1933, 1986), central to the learning process. A related concept, 'reflective practice,' was defined by Schön (1983) as the process through which professionals become aware of their implicit knowledge base and learn from experience. The concept was later developed into the notion of 'the reflective practitioner' (Cheetham & Chivers, 1998; Johns, 2009; Schön, 2017; Thompson & Thompson, 2023). Reflection grounded in professionals' own experiences has been highlighted as a critical starting point for generating new knowledge and insights (D. A. Kolb, 1984, 2007; A. Y. Kolb & Kolb, 2009; Säfström, 2017; Thompson & Thompson, 2023).

During the courses, professionals reflect upon challenges encountered in their daily work life, such as planning and prioritising goals and tasks, as well as achieving alignment between people and technology. Reflection activities are conducted both individually and in groups. As noted by Guo (2022), such practices are important for individual learning, while collective reflection, exposing professionals to multiple perspectives, further enhances the learning process. Professionals from diverse backgrounds engage in open discussions and share experiences and perspectives, thereby stimulating learning (Brott, 2023; Dissanayeke et al., 2024; Viking & Hylin, 2024).

As described, reflective practice is a common approach for professional development but it also faces criticism (Bright, 1996; Hébert, 2015; Segal, 2024). Reflection is a skill that requires training and practice for teachers as well as students. Harvey et al. (2025) emphasise that guidance is needed for effective reflection and concludes eight principles for scholarly reflective practice in learning and teaching.

General course design having an asynchronous approach

As indicated, the design and execution of the courses are grounded in Kolb's learning cycle (D. A. Kolb, 1984). Our aim is that the professionals engage in all four stages of the cycle (Figure 1), as learning is most effective when all stages are activated (D. A. Kolb, 1984, 2007).

Stage 1 (Concrete Experience) refers to the prior knowledge and experience that the professionals bring to the course. This stage is inherently addressed, through their professional experience within relevant subject areas.

In *Stage 2 (Reflective Observation)*, professionals reflect on their experiences, analyse their actions, and consider their observations in relation to prior knowledge. This reflection is stimulated by theories, methods, and examples presented in the course literature, which are used to interpret and contextualise prior experiences. These reflective activities primarily take place during engagement within the weekly modules (Figure 2).

In *Stage 3 (Abstract Conceptualisation)*, professionals develop abstract concepts and construct meaning based on their reflections and prior experiences.

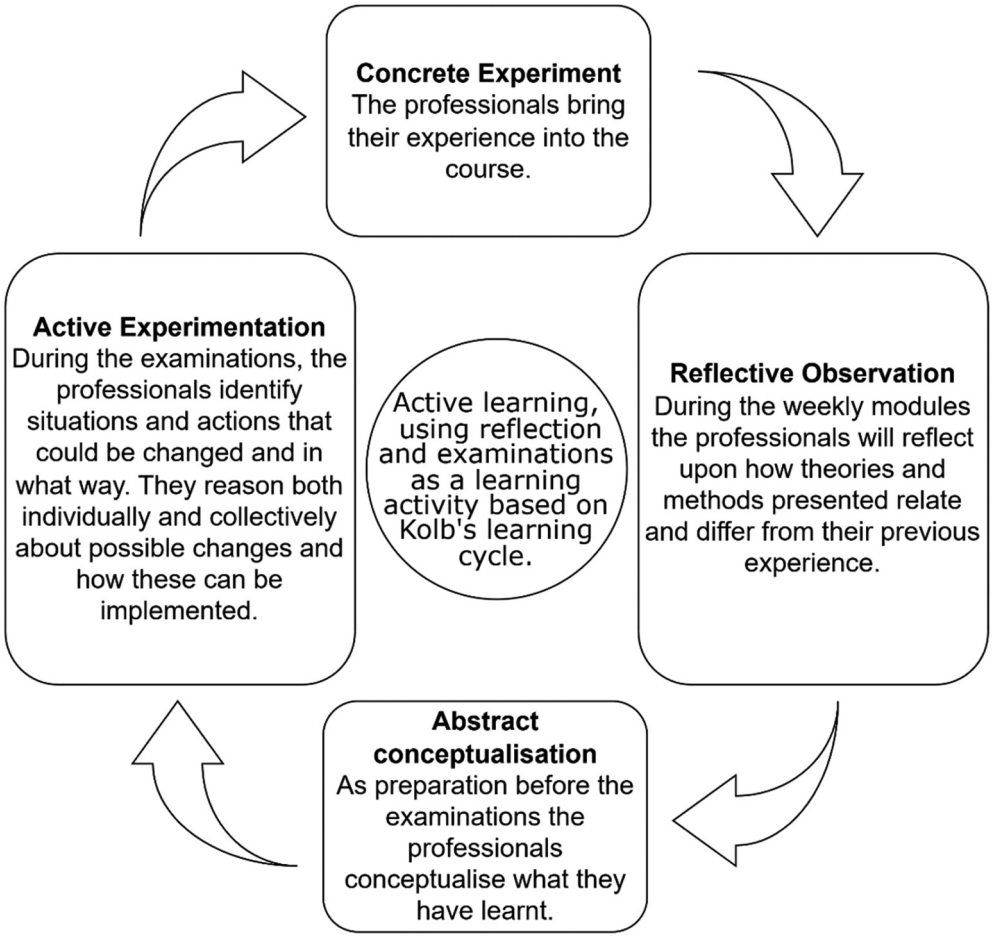


Figure 1. The application of Kolb's learning cycle in the presented courses.

Finally, *Stage 4 (Active Experimentation)* is addressed during the latter part of the course through examination activities. Professionals apply newly developed concepts to novel situations, thereby testing and evaluating their understanding in different contexts.

The examination component consists of two parts. First, in an individual written report, the professionals reflect on and analyse situations related to their professional experience. Second, in a group-based oral seminar, professionals present their own cases and discuss these, together with cases presented by others attending the same seminar, in relation to the course material. Through these activities, they reconsider how situations could have been approached differently and explore alternative courses of action.

The course design and delivery are, in addition to Kolb's learning cycle, influenced by the three aspects of online instruction proposed by Lapitan et al. (2021):

(1) Content delivery strategy
(2) Learning mechanism (asynchronous)
(3) Assessment strategy and types

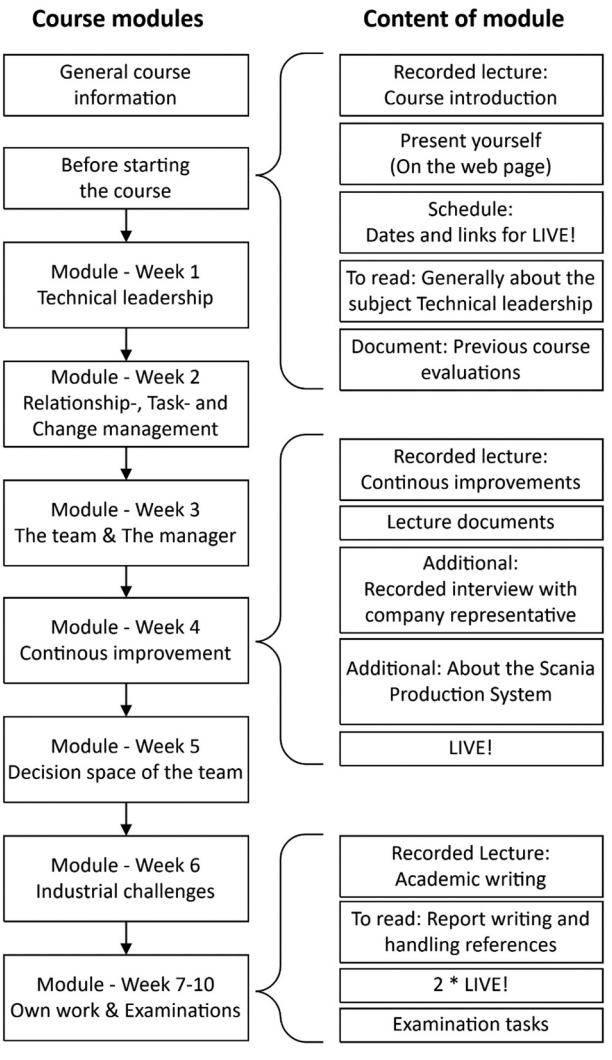


Figure 2. Examples of course modules and their content from the course technical leadership for industry in change.

Content delivery strategy

The courses are distributed over a period of 10–14 weeks and utilise the digital learning platform Canvas, where all information, documentation, and course activities are integrated. Structurally, the courses are organised into 5–8 weekly modules, each addressing a specific subtheme within the overall course topic. Each module typically includes one or two recorded lectures, along with lecture notes, required readings, and suggested supplementary materials (Figure 2). Together, these components provide the theoretical foundation for each module. The final phase of the course consists of a concluding module focused on examination activities. While the weekly modules indicate a recommended study pace, the professionals are able to adjust their progression individually, provided that examination deadlines are met.

Learning mechanism (asynchronous)

As the courses are designed for professionals working full-time, they adopt a flexible asynchronous format, enabling access to course materials at times that suit each individual's schedule (Kidron & Kali, 2024; Lapitan et al., 2021; Zeng & Luo, 2023). The importance of such flexibility is also confirmed by the participating professionals. Many of them claim it is a necessity for even enrolling in the course.

At the same time, an asynchronous approach with an excessive emphasis on flexibility, does not, by default, foster opportunities for interaction and risks limiting the professionals' ability to exchange ideas and develop a sense of learning community beyond their individual course engagement. The importance of such communities has been highlighted by Garrison (2007, 2009, 2016) and further supported by subsequent research (Martin et al., 2022; Olpak, 2022; Shea et al., 2022). Moreover, careful course design and the early inclusion of initiating activities that foster a sense of belonging are particularly important in online settings, especially for those who are 'time poor', as noted by McKenzie et al. (2024). These authors also emphasise that community-building activities should be closely aligned with course content, rather than purely social, in order to encourage participation.

To retain the benefits of an asynchronous format while also supporting interaction, dialogue, and reflection, the courses incorporate interactive elements within the overall design. The final examination is structured around reflection and interaction, and synchronous online seminars (referred to as LIVE!) are conducted periodically throughout the course using Zoom (Figure 2). The professionals are also encouraged to raise and discuss questions using the online forum at the course webpage. During the LIVE!-seminars, teachers, often together with invited guests, discuss themes and questions related to the course content. The professionals are encouraged to engage in these sessions by sharing experiences, perspectives, and reflections.

Importantly, the asynchronous structure is maintained, as all LIVE! seminars are recorded and made available via Canvas. In addition, the teachers support ongoing interaction through weekly emails (Zhao & Gaschler, 2024), in which we introduce upcoming modules, provide reminders about course activities, and respond to questions.

Assessment strategy and types

As indicated in the Introduction, this paper also addresses the role of examinations as learning activities, besides the obvious assessment role. When designed as learning activities, examinations can contribute to fostering active learning and enhancing the professionals' understanding of the relationship between theory and their professional practice.

Furthermore, the design and implementation of examinations, with related explicit grading criteria, create opportunities for the professionals to reach more advanced levels in Bloom's learning taxonomy (Bloom, 1956; Forehand, 2010) and to develop higher-order cognitive skills (such as Analyse and Evaluate, sometimes reaching Create) through reflections and discussions. In addition, engaging with higher levels of cognitive processing through reflection aligns with the third and fourth stages of Kolb's learning cycle (D. A. Kolb, 1984) (Figure 1).

Two examples of grading criteria (for Pass) for the final examination in the course *Technical Leadership in a Digital Transformation* are:

- The professional has presented a relevant, self-selected case company/organisation, which forms the basis for the reflections and discussions that are conducted.
- The professional has, based on his/her own choice, reflected upon and discussed ONE of the business dimensions; strategy, culture or employees, following the framework for digital transformation proposed by Bumann and Peter (2019).

One example of a grading criterion (for Pass with distinction) for the written report in the same course is:

- Based on both a self-selected example and relevant course material, the professional has extensively argued for the characteristics of a leader that are relevant in increased digitalisation. The professional also demonstrates an in-depth understanding of the importance of these characteristics, by highlighting several examples of negative consequences that may arise if a leader does not possess these characteristics.

The four shorter courses include two examinations; a written report and an online seminar including prepared oral presentations. The longer course additionally includes a quiz, which serves to assess the professionals' initial understanding of core theoretical concepts. In the following sections, the focus is targeted towards the two examination types, written report and online seminars, common to all five courses. These directly support a progression through the four stages of Kolb's learning cycle and foster the development of higher-order cognitive skills in accordance with Bloom's hierarchical framework for classifying educational learning objectives (Bloom, 1956).

Individual written examination

In the written reports, the professionals individually reflect on and discuss five to six cases. For each case, they are provided with a brief introduction grounded in the course literature and are asked to analyse and reflect upon the scenarios given in the cases.

For example, the professionals will be presented with a case such as:

> Your task is to select a leader that you have encountered, or use yourself as an example, and reflect on the leader's leadership signature. You are also required to explain why it is important for a leader to be clear about their signature. Additionally, provide at least one example of what may occur if the signature is unclear; that is, you should elaborate on a negative consequence of a leader lacking self-awareness regarding their behaviours and actions. (From the course*Technical Leadership in a Digital Transformation*, Autumn 2024).

When addressing such tasks, the professionals are expected to integrate their prior experiences with relevant theories from the course material, and to reflect on and justify how the leader could or should have acted. Our experience indicates that most professionals draw on situations in which they have been members of a team, reflecting on and analysing the behaviour of the team leader and its consequences.

Online seminar examinations in groups

All five courses include an online seminar as the concluding examination. Each seminar involves three professionals, who individually present a case of their own choosing and then analyse how the presented leader, individual, and/or workplace could develop based on insights gained during the course. In this process, they discuss whether and how the theories and approaches introduced in the course may support further development of both individuals and the workplace as a whole.

Each presentation lasts approximately 10–14 minutes, after which the other two professionals engage in discussion, offering reflections, brainstorming potential solutions, and analysing identified opportunities and challenges. In doing so, they actively internalise and apply the presented case, possibly finding and creating solutions to previously experienced challenges. If needed the teachers ask open questions to initiate the discussion. Once the discussion of the first case is completed, the next professional presents their case, and the process is repeated.

The evaluation of identified opportunities and proposed measures from the cases are thus conducted collaboratively, drawing on the diverse experiences and perspectives of the professionals, which strengthens the conclusions reached. It is common for the professionals to present real cases from their current workplaces, where they have already begun to implement ideas or initiate changes inspired by new insights developed through engagement with the course material and reflective activities.

Finally, the online seminars are scheduled across multiple time slots over several days, typically in the afternoons but also in the early mornings and evenings, in order to accommodate the professionals' varying schedules. Available time slots are allocated through the booking system provided within Canvas.

Discussion

Reflection and the sharing of experiences create a fertile environment in which new knowledge and theoretical understanding can develop. During the examination activities, the professionals are challenged and encouraged to reflect on how newly acquired knowledge can be applied in their daily work. Building on this foundation, they are better equipped to address emerging questions and challenges, and their capacity to adopt new technologies, as part of ongoing organisational change, is strengthened. Many indicate through their presentations that they have been encouraged to reflect, based on their own experiences, influencing their priorities, leading them to actively allocate time for reflection despite demanding schedules. Despite the relatively short duration of the courses (10–14 weeks), it is common for the professionals, during the final seminar, to describe how they have already implemented new approaches in their team meetings or project work, based on insights gained during the course. This development indicates that they have progressed through all stages of Kolb's learning cycle, including the active application of newly acquired knowledge.

The LIVE! seminars are highly appreciated, as consistently indicated in the course evaluations. However, courses with fewer active professionals tend to have significantly higher participation rates than those with larger numbers. We have also observed that, in the larger courses, discussions on the course website forum are considerably more extensive than in courses with fewer professionals. We have not seen any differences in

how many professionals access the LIVE! recordings depending on size of the course. The course design requires relatively limited effort for the delivery of the asynchronous course components, allowing teachers to allocate more time to the final seminars. So, to support a learning community, we have allocated more time for discussion and reflection during examination in the larger course, to ensure that the professionals continue to engage with one another and contribute to community building. In future courses, which may include larger groups of professionals, the number of LIVE! seminars will likely be reduced.

The examinations are deliberately designed not to assess lower levels of cognitive skills within Bloom's taxonomy (Bloom, 1956). Instead, they function as learning activities centred on reflection and discussion, creating solutions using previous experiences and new knowledge. Through the grading criteria, the professionals' reflection and discussion are evaluated based on the course material and the self-selected cases presented. Through the discussions the professionals synthesise and conclude their main learnings, finding and creating possible paths forward on previous problems faced. This approach of having discussions among three professionals during the final examination has been positively reflected in course evaluations over time. One professional expressed this in a candid but insightful way:

> I think it's very interesting to get a picture of how others feel, and it honestly feels reassuring to hear that it's not just us who struggle with these challenges (Technical Leadership for the Industrial Processes of the Future, Autumn 2022).

Although the courses were originally designed with the manufacturing industry as the primary context, it has been notable that many professionals come from the public sector. Feedback from these professionals indicates that the course content and theoretical perspectives are highly relevant also to their professional contexts.

Across all five courses, it has become evident that reflection, where prior experience is integrated with new theoretical knowledge, constitutes an effective pathway to knowledge development and the promotion of active learning. Such an approach requires the professionals to take greater responsibility for driving their own learning compared to more synchronous formats. This, combined with the demands of full-time work and everyday life, contributes to a dropout rate of approximately 20–35% during the initial weeks of the courses. At the same time, the professionals who complete the courses consistently emphasise that the asynchronous format is a prerequisite for their participation.

As previously mentioned, the courses are given in a Swedish context. However, based on our experience, having taught and discussed the structure with students and colleagues from several continents, our assessment is that the conditions for a professional who wants to study do not differ significantly. One major difference, however, is that university studies in Sweden are free of charge, which could mean that it is 'cheaper' to drop out of the course in comparison to contexts involving paid participation.

Conclusion

In light of ongoing societal and technological developments, professionals must continuously develop their knowledge in order to remain relevant and effectively respond to constant change. As a university, it is therefore essential not only to educate young students in their early twenties, but also to support professionals in the midst of their careers. In this paper, we

have examined lifelong learning initiatives aimed at working professionals, addressing two key challenges: how to design courses that can be undertaken alongside full-time work, and how reflection and active learning, grounded in Kolb's learning cycle (D. A. Kolb, 1984), can enhance professionals' knowledge development.

We have presented a general course structure based on an asynchronous approach, enabling professionals to combine lifelong learning with their everyday work. Examples and the professionals' feedback from course evaluations provide support for the effectiveness of this approach. Furthermore, we have illustrated how reflection can foster an active learning environment within the courses. Although this study focuses on lifelong learning initiatives for professionals in Sweden, the proposed course design and delivery approach may also be applicable to courses in other countries and other subject areas.

Finally, our extensive involvement in these courses over recent years has also prompted reflection on our own learning as teachers. Engaging with the professionals, their experiences, perspectives, and real-world cases, has proven highly rewarding. The opportunity for reflection and knowledge development is thus not limited to the professionals; as teachers, we have also gained valuable insights and learning.

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Ethical statement

Feedback from students are collected through anonymous course evaluations which cannot be linked to any individual student. As the study did not involve human participants, no ethical approval was required.

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